Individual Case Safety Report (ICSR)

Case ADR-2026-014 | Allopurinol-associated DRESS | Drug Safety Associate case assessment

Assessment Scope

This report covers five requested areas: ADR identification, reporting, causality assessment, severity classification, and the Drug Safety Associate (DSA) role. The DSA organizes case intake, checks the minimum validity criteria, structures and codes the event and drug information, documents the assessment and follow-up queries, and completes quality checks. Clinical judgments remain subject to medical review.

Required element	What was done	Where
ADR identification	Validity review, MedDRA event coding, differential diagnosis, and DRESS diagnostic scoring	Sections 2, 6, 12
Reporting	Narrative, structured ICSR fields, follow-up queries, reporting plan, and sign-off	Sections 1-5, 16, 17, 20
Causality assessment	WHO-UMC, Naranjo, RegiSCAR diagnostic support, and drug-by-drug review	Section 12
Severity classification	Seriousness (ICH E2A), severity grading, expectedness, and case summary	Sections 10-14
DSA role	Case intake and processing, documentation, medical-review comment, and quality checks	Sections 1, 15, 19, 20

Drug Safety Associate Role

The DSA receives and validates the case, organizes the patient, event, and medicine details, applies the appropriate coding, documents the assessment and follow-up queries, prepares the report for submission, and completes quality checks. The DSA records the evidence and uncertainty clearly and refers clinical judgments for medical review.

1. Case Administration

Field	Entry
Case ID (internal)	ADR-2026-014
Worldwide unique case ID	-
Report type	Spontaneous (healthcare professional, hospital setting)
Report version	Initial (Version 1)
Date of receipt of most recent information	-
Date of first receipt of information	-
Source country	India
Seriousness	**Serious**
Expectedness	**Expected (labelled)**
Causality (reporter / company)	Probable (WHO-UMC) / Probable (Naranjo 7)
Reporting destination	PvPI (via nearest AMC); marketing authorization holder, if applicable

2. ICSR Validity Check (minimum four criteria)

A case is a valid ICSR only if all four elements are present.

Criterion	Status	Detail
Identifiable patient	Met	58-year-old male, initials R.K., case ID ADR-2026-014
Identifiable reporter	**Pending [query]**	Reporter name and qualification not supplied; must be obtained before the case is considered valid
Suspect drug	Met	Allopurinol 300 mg OD

Criterion	Status	Detail
Adverse event	Met	DRESS syndrome with hepatic and renal involvement

Validity decision: Valid on receipt of reporter details. Obtain these first, because an unidentifiable reporter makes the report invalid for regulatory submission.

3. Patient Information

Field	Entry
Patient initials	R.K.
Age at onset	58 years
Sex	Male
Weight	72 kg
Height	6.0”
Ethnicity/region	- (relevant to HLA-B*58:01 risk)
Medical history	Type 2 diabetes mellitus (8 years), hypertension (5 years), dyslipidemia, chronic gout (diagnosed 3 weeks before treatment)
Allergies	No known drug allergies
Hepatic/renal function	No liver or kidney disease; baseline creatinine 1.0 mg/dL
Habits	Non-smoker, occasional alcohol
Pregnancy/lactation	Not applicable

4. Case Narrative

A 58-year-old male patient (72 kg) with type 2 diabetes, hypertension, dyslipidemia and newly diagnosed chronic gout was started on **allopurinol 300 mg once daily** (no dose titration) on Day 0 for urate lowering, together with colchicine 0.5 mg twice daily for flare prophylaxis. Colchicine was stopped on Day 14 after the gout flare settled. His long-term medications (metformin, amlodipine, atorvastatin) were continued unchanged.

On Day 21, he noticed mild itching on the trunk and did not seek care. On Day 25 he developed fever (101°F), a generalized maculopapular rash and facial swelling. He took paracetamol 500 mg as needed from Day 24. On Day 28 he presented to hospital with worsening rash.

On examination: temperature 101.4°F, heart rate 104/min, BP 126/80 mmHg, SpO2 97%. There was a diffuse erythematous maculopapular rash over the trunk and limbs, facial edema, tender cervical and axillary lymphadenopathy and mild hepatomegaly. There were no mucosal erosions or blisters, and Nikolsky sign was negative. Laboratory tests showed WBC 13,400/µL, eosinophils 1,400/µL (14%), ALT 210 U/L, AST 165 U/L, bilirubin 1.4 mg/dL and serum creatinine 1.9 mg/dL (baseline 1.0). Viral hepatitis markers were negative and blood culture showed no growth.

A diagnosis of **drug reaction with eosinophilia and systemic symptoms (DRESS)** was made. **Allopurinol was stopped on admission (Day 28).** He received IV fluids, topical emollients, oral antihistamines and prednisolone 1 mg/kg/day. He improved over 10 days, and liver enzymes and creatinine normalized by week 6. Rechallenge was not performed (contraindicated). The outcome was **recovered**.

The reporter assessed the event as related to allopurinol. Colchicine was considered unlikely (stopped 7 days before onset), as were the long-term drugs and paracetamol (started after the rash began).

5. Event Timeline

Day	Event
Day 0	Allopurinol 300 mg OD and colchicine 0.5 mg BD started
Day 14	Colchicine stopped (flare settled)
Day 21	Mild truncal itching (first sign); **latency 21 days**
Day 24	Paracetamol PRN started
Day 25	Fever 101°F, generalized rash, facial swelling
Day 28	Hospital presentation; **allopurinol stopped**; supportive care and prednisolone started
Day 28 + 10	Clinical improvement
Week 6	Liver enzymes and creatinine normalized

Time to onset: 21 days (first symptom) / 25 days (full syndrome) after the first dose, which fits the typical 2 to 8 week latency of DRESS.

6. ADR Identification and MedDRA Coding

Each event is coded to a MedDRA Preferred Term (PT). Verify terms against the current MedDRA version before submission.

Reported term (verbatim)	MedDRA PT	Seriousness	Outcome
DRESS syndrome	Drug reaction with eosinophilia and systemic symptoms	Serious	Recovered
Fever	Pyrexia	Serious (part of syndrome)	Recovered
Generalized itchy maculopapular rash	Rash maculo-papular; Pruritus	Serious (part of syndrome)	Recovered
Facial swelling	Face oedema	Serious (part of syndrome)	Recovered
Tender lymph nodes	Lymphadenopathy	Serious (part of syndrome)	Recovered
High eosinophils	Eosinophilia	Serious (part of syndrome)	Recovered
Raised liver enzymes with hepatomegaly	Hepatitis (or Hepatocellular injury); Hepatomegaly	Serious	Recovered
Raised creatinine	Acute kidney injury	Serious	Recovered

Primary event for the case: Drug reaction with eosinophilia and systemic symptoms. The other terms are retained as individual events for completeness.

Differential diagnosis (alternative causes considered)

Alternative	For	Against	Conclusion
Stevens-Johnson syndrome / toxic epidermal necrolysis	Drug-related delayed skin reaction	No mucosal erosions or blisters; Nikolsky sign negative	Unlikely on current findings; reassess if mucosal or blistering features appear
Viral exanthem / infectious mononucleosis	Fever, rash, and lymphadenopathy	Eosinophilia, organ injury, and close timing to a new drug support DRESS; viral PCR not done	Less likely; consider viral testing if clinically indicated [query]
Sepsis	Fever, tachycardia, and leukocytosis	Blood culture showed no growth; blood pressure and oxygen saturation were stable	Less likely on available findings; culture result alone does not exclude sepsis
Viral hepatitis	Raised ALT and AST	Viral hepatitis markers were negative	Less likely based on available marker results
Gout flare or colchicine toxicity	Gout history and a recently co-started medicine	Colchicine stopped seven days before onset; no reported marrow or gastrointestinal toxicity	Colchicine toxicity is unlikely to explain the DRESS pattern
Acute interstitial nephritis	Rise in creatinine	Occurred with the broader multisystem reaction; urinalysis not available	Not established as a separate diagnosis; assess urine findings [query]
Hypereosinophilic syndrome	Elevated eosinophils	New drug exposure preceded symptoms, with improvement after withdrawal	Less likely; reassess if eosinophilia persists or recurs

Conclusion: Based on the information available, the timing and combination of rash, fever, facial edema, lymphadenopathy, eosinophilia, and liver and kidney injury fit an allopurinol-associated adverse drug reaction (DRESS) more closely than the alternative diagnoses considered. Some alternatives remain less likely rather than definitively excluded; the follow-up tests noted in the table may help clarify them.

7. Suspect and Concomitant Drugs

Suspect drug

Field	Entry
Characterization	**Suspect**
Drug (INN)	Allopurinol
Brand/manufacturer	[to be filled]
Batch/lot, expiry	[to be filled]
Dose, route, frequency	300 mg, oral, once daily
Indication (MedDRA)	Gout / hyperuricaemia
Start / stop	Day 0 / Day 28
Duration to onset	21 to 25 days
Action taken	Drug withdrawn
Dechallenge	**Positive** (improved after withdrawal; steroids given concurrently)
Rechallenge	Not done

Concomitant drugs

Drug	Dose	Indication	Dates	Characterization
Colchicine	0.5 mg BD	Gout flare prophylaxis	Day 0 to Day 14	Concomitant (considered, not suspected)
Metformin	500 mg BD	Diabetes	6 years, ongoing	Concomitant
Amlodipine	5 mg OD	Hypertension	4 years, ongoing	Concomitant
Atorvastatin	20 mg OD	Dyslipidemia	2 years, ongoing	Concomitant
Paracetamol	500 mg PRN	Fever/pain	From Day 24	Concomitant (started after onset)

8. Tests and Procedures

Test	Result	Reference	Comment
Temperature	101.4°F		Fever
Heart rate	104/min		Mild tachycardia
BP / SpO2	126/80 mmHg / 97%		Stable
Hemoglobin	12.8 g/dL	13 to 17	Mildly low
WBC	13,400/µL	4,000 to 11,000	Leukocytosis
Eosinophils	1,400/µL (14%)	<500/µL	**Eosinophilia**
Platelets	190,000/µL	150,000 to 400,000	Normal
ALT / AST	210 / 165 U/L	<40	**Elevated**
ALP	140 U/L	44 to 147	Normal
Bilirubin	1.4 mg/dL	<1.2	Mildly high
Serum creatinine	1.9 mg/dL	0.7 to 1.3 (baseline 1.0)	**Acute kidney injury**
Uric acid	6.2 mg/dL	3.5 to 7.2	Normal
Viral hepatitis markers	Negative		Alternative cause excluded
Blood culture	No growth		Sepsis excluded

Tests not done (follow-up queries): HLA-B*58:01 genotype, atypical lymphocytes on smear, skin biopsy, HHV-6/EBV/CMV PCR, urinalysis, and serum creatinine trend between baseline and admission.

9. Treatment of the Event

Intervention	Details
Drug withdrawal	Allopurinol stopped Day 28
Supportive	IV fluids, topical emollients, oral antihistamines
Specific	Prednisolone 1 mg/kg/day (about 72 mg/day), tapered
Response	Improved over 10 days; labs normal by week 6

10. Seriousness Assessment (ICH E2A)

Criterion	Met?	Basis
Death	No	
Life-threatening	No	Stable vital signs, no organ failure
Hospitalization	**Yes**	Admitted to General Medicine ward
Disability/incapacity	No	Full recovery
Congenital anomaly	No	
Other medically important event	**Yes**	Multi-organ involvement (liver, kidney) needing systemic corticosteroids

Classification: SERIOUS. Criteria met: hospitalization and medically important event.

11. Severity Grading

Scale	Grade	Reasoning
Clinical (WHO)	**Severe**	Systemic multi-organ reaction
CTCAE v5	**Grade 3**	Hospitalization indicated; ALT/AST above 5 times ULN
Hartwig and Siegel	**Level 4b**	Treatment required and the ADR led to admission; no ICU care or permanent harm

Severity grades the intensity of the reaction, while seriousness grades the outcome or action required. They are reported separately.

12. Causality Assessment

WHO-UMC

Criterion	Finding
Plausible time relationship	Yes: onset Day 21 to 25
Unlikely due to disease or other drugs	Yes: infection and viral hepatitis excluded; other drugs excluded by timing
Response to withdrawal	Yes: positive dechallenge (confounded by steroids)
Rechallenge	Not done

Result: Probable/Likely. Not Certain, because there was no rechallenge.

Naranjo

#	Question	Score
1	Previous conclusive reports	+1
2	Event after drug given	+2
3	Improved on discontinuation	+1
4	Reappeared on rechallenge	0
5	Alternative causes	+2 (none)

#	Question	Score
6	Placebo reaction	0
7	Toxic drug level	0
8	Dose-response	0
9	Prior similar reaction	0
10	Objective evidence	+1
	Total	**7 (Probable; range 5 to 8)**

Diagnostic validation (RegiSCAR DRESS score)

Total **6** (above 5 is a definite case): lymphadenopathy +1, eosinophilia +1, rash over more than 50% of body surface (assumed) +1, suggestive rash features +1, two organs involved +2. If rash extent was under 50%, the score is 5 (probable).

Drug-by-drug causality

Drug	Causality	Reason
Allopurinol	**Probable**	Timing, characteristic reaction, positive dechallenge
Colchicine	Unlikely	Stopped 7 days before onset; not a typical DRESS trigger
Atorvastatin, amlodipine, metformin	Unlikely	Years of stable use
Paracetamol	Unlikely	Started after rash began

13. Expectedness (Listedness)

Question	Finding
Reaction in reference label?	Yes: severe hypersensitivity syndrome with rash, fever, eosinophilia, hepatitis and renal impairment is listed for allopurinol
Severity consistent with label?	Yes
Classification	**Expected (listed)**

Action: Confirm wording against the reference safety information (SmPC or package insert) of the specific brand dispensed.

14. Case Classification Summary

Parameter	Classification
Suspect drug	Allopurinol 300 mg OD
Primary event (MedDRA PT)	Drug reaction with eosinophilia and systemic symptoms
Seriousness	Serious (hospitalization, medically important event)
Severity	Severe (CTCAE Grade 3; Hartwig 4b)
Expectedness	Expected (listed)
Causality	Probable (WHO-UMC); Naranjo 7
ADR type (Rawlins-Thompson)	Type B (bizarre)
Hypersensitivity type (Gell and Coombs)	Type IV (delayed, T-cell mediated)
Preventability (Schumock-Thornton)	Potentially preventable: full 300 mg start without titration; HLA-B*58:01 not screened
Outcome	Recovered
Regulatory category	Serious, expected, spontaneous report

15. Company/Assessor Comment

The temporal relationship (onset 21 to 25 days after starting allopurinol), the clinical picture (rash, fever, facial edema, lymphadenopathy, eosinophilia, hepatitis and renal injury) and the positive dechallenge make a causal role of allopurinol probable. The syndrome is a well-documented, labelled risk of allopurinol. Colchicine, a co-started drug, was stopped 7 days before onset and is not a recognized DRESS trigger. The long-term drugs had been taken for years without incident, and paracetamol began after the rash. Infection and viral hepatitis were excluded by the tests available. The starting dose of 300 mg without titration is a recognized risk factor for allopurinol hypersensitivity, and the case is considered potentially preventable. Viral reactivation (HHV-6, EBV, CMV) has not been tested and is a possible contributor, but would not displace allopurinol as the probable cause.

16. Follow-Up Queries to Reporter

#	Query	Why it matters
1	Reporter name, qualification, institution, contact	Required for ICSR validity
2	Exact dates (start, onset, stop, recovery)	Replace relative days; confirm latency
3	Brand, manufacturer, batch and expiry of allopurinol	Product traceability and label check
4	Patient ethnicity; HLA-B*58:01 result if available	Pharmacogenomic risk factor
5	Creatinine values between baseline and admission; baseline eGFR	Renal dosing, risk assessment
6	Viral PCR, atypical lymphocytes, biopsy, urinalysis	Strengthen DRESS diagnosis and exclude alternatives
7	Exact steroid taper and any relapse during taper	Outcome and follow-up
8	Date of recovery (week 6 counted from which date?)	Outcome date accuracy
9	Who started allopurinol, and was renal function checked first?	Preventability assessment
10	Urate-lowering therapy now chosen for the patient	Safety of alternatives (cross-reactivity)

17. Regulatory Reporting Plan

Item	Plan
Report category	Serious, expected, spontaneous
Clock start (Day 0)	Date the minimum four criteria are first received by the company/AMC
Submission	PvPI AMC and, for a manufacturer, the marketing authorization holder's safety database; E2B(R3) electronic format where applicable
Timeline	Serious spontaneous cases are commonly reported within 15 calendar days; **confirm the exact timeline under PvPI/CDSCO rules and company SOP**
Periodic reporting	Include in PBRER/PSUR line listing as a serious listed case
Follow-up	Submit follow-up after the steroid taper, or when query responses arrive
Signal management	Add to allopurinol DRESS case count; monitor trend against label

18. Risk Minimization and Clinical Recommendations

1. Stop allopurinol permanently and record "Allopurinol: DRESS, severe, avoid" in the patient's allergy/ADR record.
2. Give the patient an ADR alert card and counsel them.
3. Continue the corticosteroid taper slowly over 6 to 8 weeks and watch for relapse.
4. Monitor liver function, renal function and eosinophil count at follow-up.
5. Choose any alternative urate-lowering drug carefully (febuxostat with caution because of possible cross-reactivity; uricosurics if renal function allows).
6. For future patients: start allopurinol at 50 to 100 mg/day, titrate gradually, and consider HLA-B*58:01 screening in high-risk populations.

19. Quality Control Checklist

Check	Status
Four minimum criteria present	Pending reporter details
Narrative consistent with structured data	Done
Dates and latency consistent	Relative days only; exact dates pending
Events coded to MedDRA PTs	Done (verify version)
Seriousness, severity and expectedness assigned	Done
Causality documented with method	Done
Concomitant drugs listed with dates	Done
Duplicate case check	Search safety database for the same patient and event; [to be filled]
Follow-up queries issued	Listed in section 16
Medical review	[to be filled]

20. Sign-Off

Role	Name	Date	Signature
Drug Safety Associate (case processor)	[to be filled]	[to be filled]	_____
Medical reviewer	[to be filled]	[to be filled]	_____
QC reviewer	[to be filled]	[to be filled]	_____

Patient and reporter identities are confidential and handled under applicable data protection requirements.